

Study [ref]	Design	n	Risk of bias	Access outcomes (technology in use)						Learning outcomes (unaided)		
				Compre- hension	Reading effic.	Text length	Transcr. accuracy	Text quality	Engage- ment	Unaided literacy	Curric. attain.	Braille acquis.
Speech-to-text												
Raskind & Higgins (1999) [29,30]	NRS	52	Higher							▲		
MacArthur & Cavalier (2004) [31]	NRS	21	Lower					▲				
Kraft (2023) [32]	NRS	16	Lower				▲	◀▶				
Almgren Bäck et al. (2024) [33]	SCED	8	Higher			▲	◀▶	◀▶				
Fälth et al. (2025) [34]	SCED	7	Higher			◀▶		◀▶				
Sand et al. (2025) [35]	SCED	4	Higher			▲		▲				
Text-to-speech / mixed												
Bhola (2022) [36]	RCT	20	Higher								▲	
Grunér et al. (2018) [37]	RCT	49	Lower	◀▶	▲							
Svensson et al. (2021) [38]	NRS	149	Higher							◀▶		
Keelor et al. (2023) [39]	NRS	29	Lower	▲								
Ebajay & Malabo (2026) [40]	NRS	20	Higher								▲	
Chen et al. (2026) [41]	NRS	38	Higher	◀▶	▲							
Bonifacci et al. (2022) [42]	NRS	20	Lower	▲					▲			
Silvestri et al. (2022) [43]	NRS	94	Lower	◀▶								
Young et al. (2019) [44]	SCED	4	Lower	▲						◀▶		
Sulaimon & Schaefer (2023) [45]	SCED	2	Lower	▲								
Schmitt et al. (2019) [46]	SCED	4	Lower	▲	▲							
Alqahtani (2023) [47]	SCED	3	Lower	◀▶	▲							
Nuraini Herawati et al. (2022) [48]	SCED	1	Higher							▲		
Svensson et al. (2026) [49]	SCED	8	Higher	▲						◀▶		
Moorman et al. (2010) [50]	SCED	2	NA	▲	▲							
Meyer & Bouck (2014) [51]	SCED	3	Lower	◀▶	◀▶							
Meyer & Bouck (2017) [52]	SCED	4	NA	◀▶								
Brunow & Cullen (2021) [53]	SCED	4	Lower	▼								
Fälth et al. (2025) [54]	SCED	5	Higher	◀▶					◀▶			
Adaptive Braille learning												
McCarthy et al. (2016) [55]	SCED	10	Higher									▲

▲ favours technology ▼ favours comparator ◄► no clear difference, mixed or conflicting (direction coded without statistical significance).
 Marker size: n ≥ 50, 10–49, < 10 learners with disabilities. Fill: black = lower risk of bias (RoB 2 some concerns; ROBINS-I moderate;
 WWC meets with reservations); grey = higher risk (high/serious; does not meet WWC); white = not assessable (abstract only). NRS = non-randomised or
 within-participant; SCED = single-case experimental design.